

AI-Powered Medical Report Interpretation System

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Healthcare institutions generate vast volumes of medical reports daily, including laboratory results, diagnostic scans, and clinical documents. Manual interpretation of these reports is time-consuming and error-prone, particularly when reports contain complex terminology or critical abnormal values. Existing digital health tools often lack structured mechanisms for patient record accountability, leaving significant gaps in data transparency and clinical governance.

This project addresses the absence of auditable, role-aware patient record management in lightweight clinical support systems. While basic electronic health record platforms offer data storage, few incorporate synchronized change tracking with clinical AI assistance tailored for doctor use in resource-constrained or institutional academic environments.

The proposed AI-Powered Medical Report Interpretation System is a secure web-based platform designed to assist qualified doctors in organizing, analyzing, and interpreting patient medical reports efficiently. The system incorporates role-based access control with medical license verification, ensuring that only authorized personnel interact with sensitive clinical data. A core module handles structured patient profile and clinical record management, featuring a robust audit log that captures every modification — recording the responsible user, the nature of the change, and the precise timestamp. This aligns the system with real-world healthcare governance and accountability standards. An integrated AI chatbot further supports doctors by answering clinical queries using professional medical language, while an AI analysis engine extracts key values, flags abnormal results, and generates structured report summaries with appropriate disclaimers.

Technologies employed include Python with Flask or Django for backend development, React.js for the frontend interface, PostgreSQL for relational data storage, and large language model APIs for AI-driven report interpretation and chatbot functionality.

Expected outcomes include reduced report interpretation time, improved clinical data traceability, and a scalable framework adaptable to hospital information systems. Socially, the system promotes safer and more efficient healthcare delivery. Commercially, it offers a foundation for certified clinical decision-support software.

Keywords: Patient Record Management, Audit Logging, Change Tracking, Clinical Data Governance, Role-Based Access Control, Healthcare Information Systems

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LIST OF ABBREVIATIONS

Abbreviation	Full form
XAI	Explainable Artificial Intelligence
CRUD	Create, Read, Update, Delete
EHR	Electronic Health Record
FHIR	Fast Healthcare Interoperability Resources
GDPR	General Data Protection Regulation
JWT	Json Web Token
HIPAA	Health Insurance Portability and Accountability Act
API	Application Programming Interface
UI	User Interface
GDPR	General Data Protection Regulation
RBAC	Role- Based Access Control
REST	Representational State Transfer
LLM	Large Language Model

LIST OF APPENDICES

Appendix	Title	Description
Appendix A	Survey Instruments	Questionnaires and data collection forms used for requirements gathering and user feedback
Appendix B	Survey Instruments	Comprehensive system architecture, database schema (ERD), and component interaction diagrams
Appendix C	Preliminary UI Mockups	Initial wireframes and interface designs for the patient profile, audit log, and dashboard screens.
Appendix D	Risk Analysis	Identified project risks, probability ratings, impact assessments, and mitigation strategies.
Appendix E	Ethical Documents	Ethics clearance forms, data privacy declarations, and informed consent templates used in the research.

INTRODUCTION

1.1 Background and Context

Modern healthcare systems generate an enormous amount of medical data on a daily basis. Hospitals, diagnostic laboratories, and healthcare institutions continuously produce large volumes of patient-related information such as laboratory test results, radiology scans, pathology reports, prescriptions, and clinical notes. These medical reports play a crucial role in the diagnosis, monitoring, and treatment of patients. Doctors rely heavily on these reports to identify diseases, evaluate patient conditions, and determine appropriate medical treatments. Therefore, the ability to accurately analyze and interpret medical reports is essential for ensuring high-quality healthcare services and improving patient outcomes.

However, the rapid growth of healthcare data has created significant challenges for healthcare professionals. The increasing number of diagnostic tests and advanced medical technologies has led to a substantial rise in the amount of data generated within hospitals. As a result, doctors must review and analyze large numbers of medical reports within limited timeframes. Many medical reports contain complex clinical terminology, numerical test values, and detailed diagnostic observations that require expert interpretation. When doctors are required to manually review large volumes of such reports, the process becomes time-consuming and cognitively demanding.

This issue affects several stakeholders within the healthcare ecosystem, including doctors, hospital administrators, and patients. Doctors experience increased workloads when reviewing medical reports manually, especially in busy healthcare environments where the number of patient cases is high. The pressure to analyze reports quickly may increase the risk of overlooking critical information or misinterpreting abnormal values. Hospital administrators are responsible for ensuring that healthcare services are delivered efficiently while maintaining the accuracy and reliability of patient records. Meanwhile, patients depend on

doctors to interpret their medical reports correctly in order to receive accurate diagnoses and appropriate treatment. Therefore, improving the efficiency and reliability of medical report interpretation is a critical requirement in modern healthcare systems.

Globally, the volume of healthcare data has been growing at an unprecedented rate. According to various research studies, healthcare data is expected to grow significantly due to the increasing use of digital health technologies, electronic health records, and advanced diagnostic equipment. Reports indicate that healthcare generates one of the fastest-growing data volumes among all industries. The adoption of electronic medical systems and digital imaging technologies has further accelerated the production of medical information. In large hospitals, thousands of medical reports are generated every day, making it increasingly difficult for healthcare professionals to manually analyze and manage all the information effectively.

To manage this growing volume of data, many healthcare institutions have adopted digital platforms such as Electronic Health Record (EHR) systems and Hospital Information Systems (HIS)**.** These systems allow hospitals to store, manage, and retrieve patient information electronically. They provide centralized access to patient records, enabling doctors and healthcare staff to access medical histories, laboratory results, and clinical notes more easily. The use of EHR systems has significantly improved the organization and accessibility of healthcare data compared to traditional paper-based record systems.

Despite these advantages, most existing healthcare information systems primarily focus on data storage and administrative management rather than intelligent analysis of medical reports. Doctors still need to manually read and interpret the contents of each report in order to understand the medical findings. These systems generally do not provide automated explanations of complex medical terms or highlight abnormal test results that may require urgent attention. As a result, doctors must spend considerable time reviewing medical documents, which may reduce overall efficiency in healthcare delivery.

Another limitation of many existing systems is the lack of effective mechanisms for tracking changes made to patient records. In real hospital environments, patient records are frequently updated as new test results become available or as doctors add clinical observations. Without proper change tracking mechanisms, it

can be difficult to determine who modified a particular record, what information was changed, and when the change occurred. This lack of transparency can create challenges for maintaining accountability and compliance with healthcare regulations. Accurate audit trails are essential for ensuring data integrity, preventing unauthorized modifications, and maintaining trust in healthcare information systems.

With the advancement of digital technologies, Artificial Intelligence (AI) has emerged as a promising solution for addressing some of these challenges in healthcare data management. AI technologies such as machine learning and Natural Language Processing (NLP) can analyze large volumes of medical data and extract meaningful insights from complex clinical documents. These technologies can help identify key medical terms, summarize medical reports, and detect abnormal values in laboratory results. AI-based systems can therefore assist doctors in understanding complex medical information more quickly and efficiently.

The integration of AI into healthcare systems has the potential to significantly improve the efficiency of medical data analysis. By providing automated support for interpreting medical reports, AI can reduce the cognitive burden on healthcare professionals and help them focus more on clinical decision-making and patient care. However, it is important that such systems are designed to support doctors rather than replace their professional judgment. AI tools should provide explanations and insights while ensuring that final medical decisions remain under the control of qualified healthcare professionals.

To address the challenges associated with medical report interpretation and patient record management, this research proposes the development of an AI-Powered Medical Report Interpretation System. The proposed system is designed as a secure digital platform that assists qualified doctors in understanding and analyzing patient medical reports more efficiently. The system integrates several important components, including secure doctor authentication, patient profile management, clinical record management, AI-based report analysis, and a doctor-oriented AI chatbot.

A key component of the proposed system is the Patient Profile and Clinical Record Management module with Change Tracking**.** This module allows doctors to create structured patient profiles, upload medical reports, and maintain clinical notes

within a secure digital environment. Unlike basic record management systems that simply store patient data, this module includes a comprehensive audit logging mechanism that tracks every modification made to patient records. The system records details such as the user who performed the modification, the specific data that was changed, and the exact time when the change occurred. This feature ensures transparency, accountability, and traceability in patient record management.

In addition to record management, the system includes an AI-powered engine capable of analyzing uploaded medical reports. The AI component extracts key medical terms and identifies abnormal test results using predefined medical rules and analytical techniques. It then generates structured explanations that help doctors quickly understand the key findings in the report. Furthermore, the system includes a doctor-oriented AI chatbot that allows doctors to ask questions about medical reports using professional medical language. The chatbot provides explanations and summaries while clearly indicating that it does not replace clinical decision-making.

The proposed system contributes to improving healthcare services by enhancing the efficiency of medical report interpretation and strengthening patient record management practices. By integrating AI-based analysis with transparent audit mechanisms, the system provides a more reliable and accountable healthcare information management framework.

This research aligns with the United Nations Sustainable Development Goal (SDG) 3: Good Health and Well-Being, which aims to ensure healthy lives and promote well-being for people of all ages. The use of intelligent digital technologies to support healthcare professionals contributes to improving the quality, accessibility, and efficiency of healthcare services.

Furthermore, the project aligns with the Health Informatics and Artificial Intelligence research cluster**,** which focuses on applying advanced computing technologies to solve problems in healthcare data management, medical information systems, and clinical decision support. By integrating AI with secure patient record management and change tracking mechanisms, this research demonstrates how modern information technologies can be used to enhance healthcare information systems and support medical professionals in delivering better patient care.

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1.2 Literature Review

A review of recent scholarly literature reveals several important insights relevant to the design of this module.

1.2.1 Electronic Health Record Systems

Bates et al. (2003) demonstrated in a landmark study that well-designed EHR systems with structured data entry and audit capabilities reduced adverse drug events by 55% in hospital settings. More recently, Cresswell et al. (2017) reviewed the implementation of EHR systems across 14 countries and found that governance features — particularly audit logs and role-based access — were the most consistently cited factors in successful deployments. These findings establish a strong evidence base for the inclusion of audit logging as a core system feature rather than an optional add-on.

1.2.2 Audit Logging and Healthcare Data Governance

Rind fleisch (1997) identified four primary threats to medical record privacy: curiosity, financial gain, malice, and authorized access for unauthorized purposes. Audit logs directly mitigate all four threats by creating a deterrent effect and enabling forensic investigation after incidents. The HL7 FHIR Audit Event resource provides a standardized schema for capturing system activity logs in healthcare applications, including actor identification, action type, timestamp, and outcome. This proposal draws directly on this standard in designing the audit log schema.

1.2.3 Role-Based Access Control

(1992) formally defined RBAC as an access control mechanism that assigns permissions to roles rather than to individual users, significantly simplifying administration in large organizations. In

clinical environments, (2012) found that RBAC implementations must account for the dynamic nature of clinical roles, where a doctor may have different access rights depending on their current context — for example, whether they are treating a patient directly or consulting on a case.

1.2.4 Data Versioning and Temporal Records

Jensen and Snodgrass (1999) introduced the concept of bitemporal databases, which maintain both valid time and transaction time for all records, enabling complete historical reconstruction of any data state. While full bitemporal implementation adds complexity, Snodgrass (1995) demonstrated that even simple append-only audit tables provide substantial benefits for legal defensibility and data recovery.

1.2.5 Existing Systems and Their Limitations

A review of comparable student research projects and open-source clinical management tools (OpenMRS, FreeMED, GNU Health) reveals that most systems either lack formal audit capabilities entirely or implement only basic timestamps without field-level change tracking .Furthermore, none of the reviewed student-level systems implement soft deletion with recovery, concurrent edit detection, or data completeness scoring — all of which are standard features in enterprise EHR systems such as Epic and Cerner.

1.3 Research Gap

Although healthcare information systems and AI-based medical analysis tools have evolved significantly, several important gaps remain in existing technologies. Most current hospital information systems focus primarily on storing patient data and facilitating administrative tasks rather than assisting doctors in interpreting complex medical reports.

Doctors still rely heavily on manual interpretation of medical reports, which can be time-consuming and may increase the likelihood of human error. Existing AI tools for medical analysis often operate as standalone applications rather than being integrated into hospital record management systems. As a result, AI-generated insights are not always directly connected to patient records.

Another significant limitation of many existing systems is the lack of comprehensive change tracking and audit mechanisms. When patient records are modified, it is important to maintain a clear history of changes in order to ensure accountability and data integrity. However, many systems allow records to be edited without maintaining detailed information about the modifications that occurred.

Furthermore, traditional student-level record management systems typically rely on simple CRUD (Create, Read, Update, Delete) operations, where data is overwritten without maintaining historical versions. This approach does not meet the governance standards required for professional healthcare information systems.

The proposed AI-Powered Medical Report Interpretation System aims to address these limitations by integrating AI-based report interpretation with structured patient record management and advanced change tracking capabilities. The system introduces an auditable patient record management framework, where every modification to patient data is recorded along with details about the user, the specific changes made, and the timestamp of the modification.

This approach ensures transparency, traceability, and accountability in patient record management while maintaining data consistency across patient profiles, clinical notes, medical reports, and AI-generated outputs.

The novelty of this research lies in combining AI-driven medical report interpretation with governance-level patient record management, creating a secure and intelligent healthcare platform that supports doctors without replacing their clinical judgment.

2.OBJECTIVE

2.1 Main Objective

To design and implement a governance-compliant Patient Profile and Clinical Record Management module featuring an immutable audit log and role-based access control, enabling transparent, accountable, and traceable management of patient clinical data within the AI-Powered Medical Report Interpretation System.

2.2 Specific Objectives

1. Design a normalized relational database schema for patient profiles, clinical notes, medical report metadata, and audit logs using PostgreSQL.
2. Implement a Role-Based Access Control (RBAC) system with three distinct roles — Junior Doctor, Consultant, and Administrator — each with differentiated data access permissions.
3. Develop an immutable, append-only audit log that automatically captures all patient data modifications, including the field changed, old value, new value, responsible clinician, and precise timestamp.
4. Implement soft deletion with full record recovery capability, ensuring that no patient data is permanently destroyed.
5. Build a clinical timeline view that presents a chronological, aggregated history of all patient record changes and clinical events.
6. Implement a data completeness scoring mechanism that evaluates and displays the percentage of required patient profile fields that have been populated.

3. METHODOLOGY

3.1 Research Approach

This research follows the Design Science Research (DSR) methodology, as defined by Hevner et al. (2004), which is the most appropriate paradigm for research that produces an IT artefact — in this case, a working software module — as its primary output [12]. DSR involves iterative cycles of design, build, and evaluation, making it well-suited to software engineering research projects.

3.2 Technical Methods and Algorithms

Change Detection Algorithm

A differential comparison algorithm will be implemented at the service layer. Before persisting any update to a patient record, the system retrieves the current state from the database and computes a field-by-field diff. For each field where the old value differs from the new value, an audit log entry is automatically generated. This process is transparent to the clinician and requires no additional user action.

Soft Deletion Pattern

Instead of issuing SQL DELETE statements, all deletions will set a deleted_at timestamp and a deleted_by foreign key on the relevant record. A database view will exclude soft-deleted records from all standard queries, while a separate administrative query provides access to the archive for recovery purposes.

Data Completeness Scoring

A weighted scoring algorithm will evaluate each patient profile against a checklist of required and optional fields. Required fields (date of birth, blood type, emergency contact) carry higher weights than optional fields. The resulting score, expressed as a percentage, is displayed on the patient dashboard to prompt clinicians to complete incomplete records.

3.3 Tools and Platforms with justification

Layer	Technology	Justification
Frontend	React js + Tailwind CSS	Component-based UI, fast rendering, widely adopted in industry
Backend	Node.js + Express.js	Lightweight, asynchronous, ideal for REST APIs
Database	Super Base	ACID-compliant, relational, supports complex queries and audit patterns
Authentication	JWT (JSON Web Tokens)	Stateless, secure, scalable token-based authentication
Testing	Jest + Supertest	Unit and integration testing for backend API endpoints
Version Control	Git + GitHub	Collaborative development and code versioning
Project Management	Trello / Jira	Task tracking and sprint management

3.4 Data Collection and Ethical Considerations

No real patient data will be used in this system. All testing will be conducted using synthetically generated patient records created specifically for evaluation purposes. The system will not be deployed in a live clinical environment. Data privacy by design principles will be followed, including field-level access restrictions, encrypted storage of sensitive fields, and strict password hashing using by crypt. The research has no ethical concerns regarding human subjects.

3.5 Validation Strategy and Performance Metrics

- Unit testing of all audit log trigger functions to verify correct field-level change capture.
- Integration testing of RBAC rules to confirm that access violations are correctly rejected.
- Load testing to ensure the system handles at least 100 concurrent users without performance degradation.
- Manual review of audit logs against a known sequence of changes to validate accuracy and completeness.
- Comparison of feature set against three comparable student projects to demonstrate novelty.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

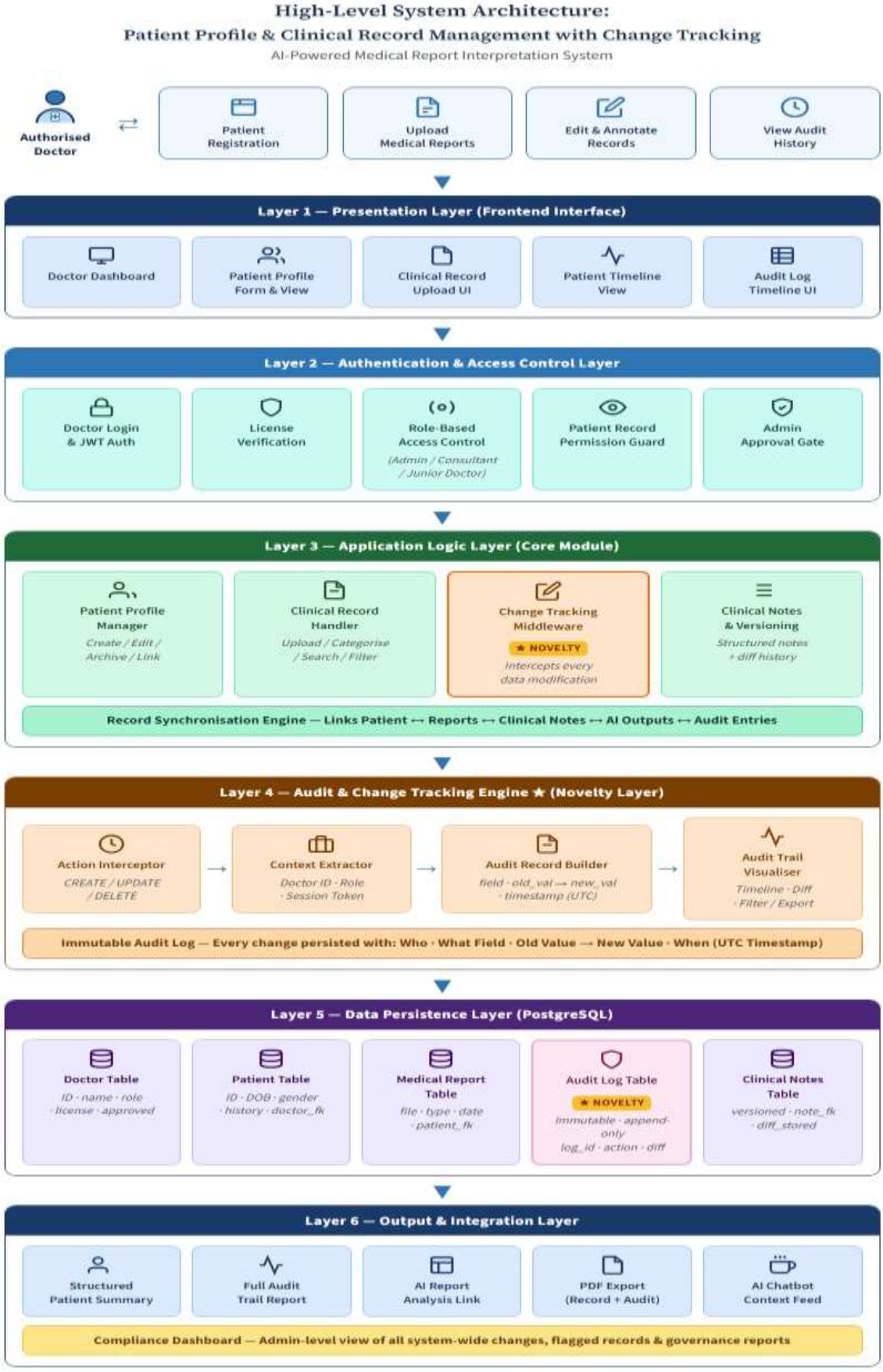
4.1 Overview of the Proposed Solution and its components

The AI-Powered Medical Report Interpretation System is designed as a secure and modular digital platform that assists qualified doctors in analyzing and understanding patient medical reports. The system integrates artificial intelligence, structured patient data management, and secure authentication mechanisms to improve efficiency and transparency in healthcare data handling.

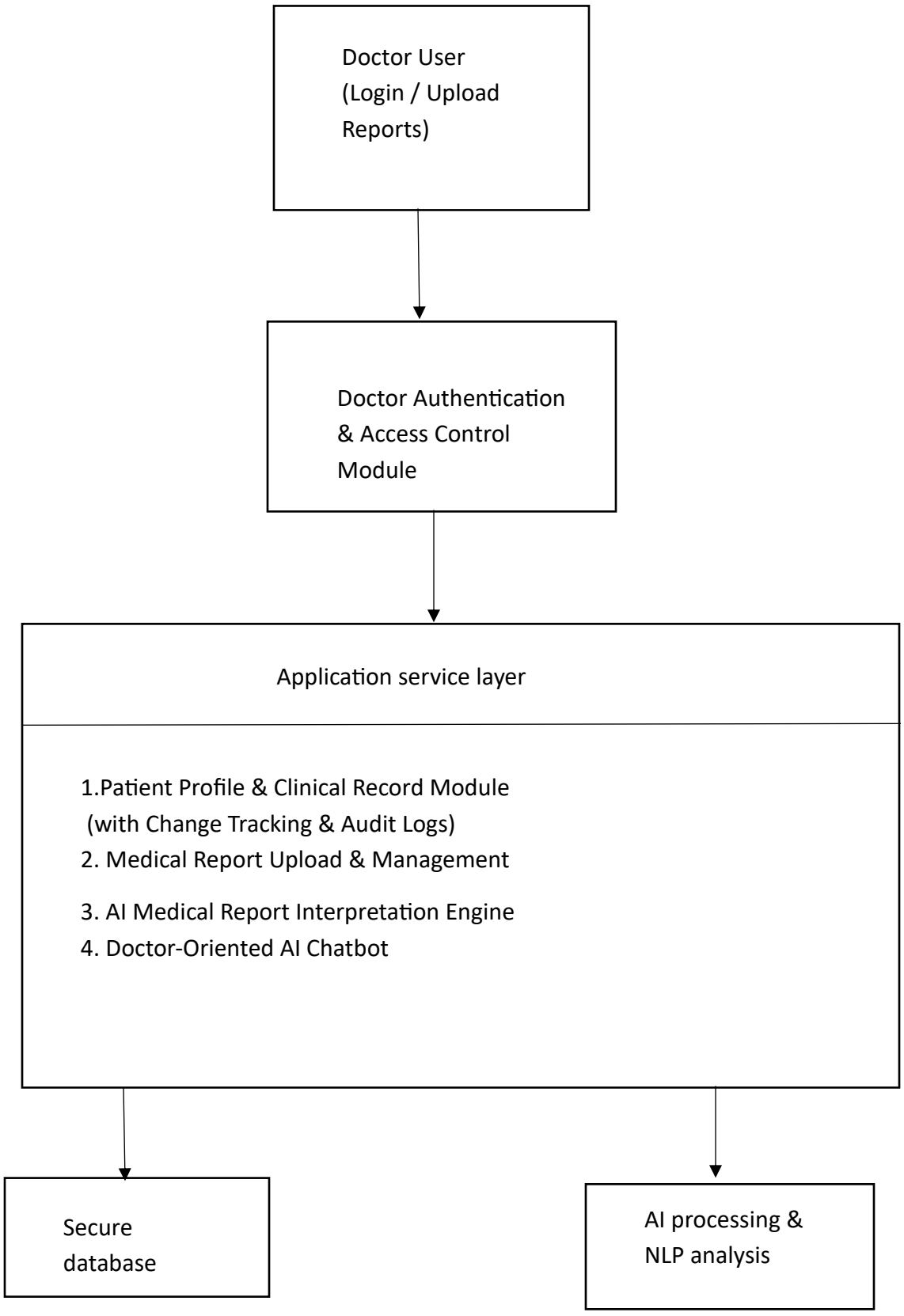
The architecture follows a multi-layered modular design, allowing different components to work together while maintaining security, scalability, and data consistency. Each module performs a specific function within the system, such as authentication, patient record management, AI analysis, and report interaction through an AI chatbot.

The main components of the system include:

1. Doctor Authentication and Access Control Module
2. Patient Profile & Clinical Record Management Module with Change Tracking
3. AI Report Interpretation Engine
4. Doctor-Oriented AI Chatbot


High-Level System Architecture:
Patient Profile & Clinical Record Management with Change Tracking
AI-Powered Medical Report Interpretation System
Authorised Doctor
Patient Registration
Upload Medical Reports
Edit & Annotate Records
View Audit History
Layer 1 — Presentation Layer (Frontend Interface)
Doctor Dashboard
Patient Profile Form & View
Clinical Record Upload UI
Patient Timeline View
Audit Log Timeline UI
Layer 2 — Authentication & Access Control Layer
Doctor Login & JWT Auth
License Verification
Role-Based Access Control
(Admin / Consultant / Junior Doctor)
Patient Record Permission Guard
Admin Approval Gate
Layer 3 — Application Logic Layer (Core Module)
Patient Profile Manager
Create / Edit / Archive / Link
Clinical Record Handler
Upload / Categorise / Search / Filter
Change Tracking Middleware
★ NOVELTY
Intercepts every data modification
Clinical Notes & Versioning
Structured notes + diff history
Record Synchronisation Engine — Links Patient ↔ Reports ↔ Clinical Notes ↔ AI Outputs ↔ Audit Entries
Layer 4 — Audit & Change Tracking Engine ★ (Novelty Layer)
Action Interceptor
CREATE / UPDATE / DELETE
Context Extractor
Doctor ID · Role · Session Token
Audit Record Builder
field · old_val → new_val · timestamp (UTC)
Audit Trail Visualiser
Timeline · Diff · Filter / Export
Immutable Audit Log — Every change persisted with: Who · What Field · Old Value → New Value · When (UTC Timestamp)
Layer 5 — Data Persistence Layer (PostgreSQL)
Doctor Table
ID · name · role · license · approved
Patient Table
ID · DOB · gender · history · doctor_fk
Medical Report Table
file · type · date · patient_fk
Audit Log Table
★ NOVELTY
immutable · append-only
log_id · action · diff
Clinical Notes Table
versioned · note_fk · diff_stored
Layer 6 — Output & Integration Layer
Structured Patient Summary
Full Audit Trail Report
AI Report Analysis Link
PDF Export (Record + Audit)
AI Chatbot Context Feed
Compliance Dashboard — Admin-level view of all system-wide changes, flagged records & governance reports

4.2 System Diagram of component



4.3 Individual Components and Their Parts

1. Doctor Authentication & Access Control Module

This module ensures that only authorized medical professionals can access the system.

Key Parts

- Doctor login system
- Medical license verification
- Role-based access control
- Administrator approval system
- Session management and security controls

Purpose

To maintain ethical and secure usage of the system by restricting access to verified healthcare professionals.

2. Patient Profile & Clinical Record Management Module and Audit Logging & Change Tracking System

This module manages structured patient information and clinical records within the system.

Key Parts

- Patient profile creation
- Patient demographic information storage
- Medical history management
- Clinical notes storage
- Medical report linking
- Doctor-patient relationship management
- Change tracking mechanism
- Audit log generation

Key Innovation

Unlike basic data management systems, this module introduces audit logging and change tracking, which records:

- Who modified the patient record
- What information was changed
- When the change occurred

This ensures data transparency, accountability, and traceability, which are essential in professional healthcare environments.

Key Features

- Change history logs
- User activity tracking
- Record modification timestamps
- Version tracking for records

This feature ensures legal compliance and transparency in medical data management

3. Medical Report Upload and Management Module

This module allows doctors to upload and manage medical documents.

Key Parts

- Upload laboratory reports
- Upload radiology scans
- Upload pathology documents
- File format validation (PDF, images, etc.)
- Secure report storage
- Linking reports to patient profiles

4. AI Medical Report Interpretation Engine

This component analyzes uploaded medical reports using AI techniques.

Key Parts

- Medical text extraction
- NLP-based medical term identification
- Abnormal value detection
- Report summarization
- Structured explanation generation
- Safety rule-based medical validation

Purpose

To help doctors quickly understand important findings in medical reports without replacing professional medical judgment.

5. Doctor-Oriented AI Chatbot

The AI chatbot provides interactive support for doctors.

Key Parts

- Natural language query interface
- Medical report explanation
- Context-based question answering
- Report summary discussion
- Safety disclaimers

The chatbot assists doctors by explaining report details but does not provide diagnoses or treatment decisions**.**

4.4 Component Interactions

The interaction between components occurs through a structured workflow.

1. The doctor logs into the system using secure authentication.
2. The doctor creates or accesses a patient profile.
3. Medical reports are uploaded and linked to the patient record**.**
4. The AI engine processes the uploaded report and extracts medical insights.
5. AI explanations and summaries are stored and linked to the report.
6. Doctors may interact with the AI chatbot to ask questions about the report.
7. Any modifications to patient data are recorded by the audit logging system.
8. All information is stored securely in the centralized database.

4.5 Data Flow Explanation

The system follows a secure and structured data flow.

Step 1 – Authentication

Doctor credentials are verified by the authentication module.

Step 2 – Patient Data Access

Once authenticated, the doctor can access or create patient profiles.

Step 3 – Report Upload

Medical reports are uploaded and linked to the patient record.

Step 4 – AI Processing

The AI engine analyzes the uploaded report and extracts key information.

Step 5 – AI Explanation Generation

The system produces structured explanations highlighting:

- Key medical terms
- Abnormal test values
- Important report findings

Step 6 – Storage and Logging All data and modifications are stored in the secure database while change logs track every modification.

4.6 Scalability and Security Considerations

Scalability

The system architecture is designed to scale efficiently as hospital data grows.

Key scalability strategies include:

- Modular architecture for independent component scaling
- Cloud-based storage for large medical datasets
- AI processing services that can scale based on workload

- Database indexing for faster patient record retrieval

Security Considerations

Healthcare data requires strict security protections.

The system implements several security mechanisms:

1. Secure Authentication

Only verified doctors can access the system.

2. Role-Based Access Control

Different access levels for doctors, consultants, and administrators.

3. Data Encryption

Patient records and reports are encrypted during storage and transmission.

4. Audit Logging

All modifications to patient data are recorded for accountability.

5. Privacy Protection

Sensitive medical information is protected according to healthcare data governance standards.

USER REQUIREMENTS

5.1 Stakeholder identification

- Junior Doctors — create and view patient profiles; add clinical notes; view AI-generated report summaries.
- Consultant Doctors — all junior doctor permissions plus the ability to edit existing records and access complete audit trails.
- System Administrators — full system access including user management, role assignment, and access to archived (soft-deleted) records.

5.2 Functional Requirements

ID	Requirement	Priority
F1	The system shall allow authorized doctors to create new patient profiles.	High
F2	The system shall allow doctors to view full patient profiles including medical history.	High
F3	The system shall allow consultants and admins to update patient profile fields.	High
F4	The system shall automatically generate an audit log entry for every field-level change.	High
F5	The system shall support soft deletion — records are archived, not permanently deleted.	High
F6	The system shall allow administrators to recover soft-deleted patient records.	Medium
F7	The system shall display a clinical timeline view of all patient record events	Medium
F8	The system shall calculate and display a data completeness score for each patient profile.	Medium
F9	The system shall restrict access to patient records based on the doctor's assigned role.	High
F10	The system shall allow doctors to add and view clinical consultation notes per patient.	High
F11	The system shall link uploaded medical reports to the corresponding patient profile.	High
F12	The system shall synchronize audit records with AI-generated outputs per patient.	Medium

5.3 Non-Functional Requirements

ID	Requirement	Category
NF 1	All API responses shall be returned within 2 seconds under normal load.	Performance
NF 2	The system shall support at least 100 concurrent users without degradation.	Scalability
NF 3	All audit log records shall be immutable — no update or delete operations permitted.	Data Integrity
NF 4	All data transmissions shall be encrypted using HTTPS/TLS 1.2 or higher.	Security
NF 5	Passwords shall be hashed using crypt with a minimum cost factor of 12.	Security
NF 6	The UI shall be responsive and accessible on desktop and tablet devices	Usability
NF 7	The system shall maintain 99.5% uptime during the evaluation period.	Reliability
NF 8	All sensitive fields shall be encrypted at rest using AES-256.	Security

6.COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market consists of private hospitals and multi-specialty clinics in Sri Lanka and the South Asian region that are seeking to digitise their clinical record management without the cost and complexity of enterprise EHR systems. The customer persona is a Hospital IT Manager or Chief Medical Officer at a 50–200 bed private hospital who needs a cost-effective, compliant, and user-friendly clinical data management solution.

6.2 Value Proposition

The system offers governance-compliant patient record management with built-in audit trails at a fraction of the cost of enterprise systems like Epic or Cerner. It provides immediate value through improved accountability, reduced risk of data loss, and readiness for regulatory inspection — without requiring expensive implementation consultants or long deployment timelines.

6.4 Competitive Advantage

- Field-level immutable audit logging is not offered by any comparable Sri Lankan clinical software at this price point.
- Built specifically for Sri Lankan healthcare context, including local regulatory awareness.
- Modular architecture allows the patient management module to be licensed independently or as part of the full system.
- Potential to integrate with the Ministry of Health's national eHealth infrastructure.

6.5 Intellectual Property Considerations

The system will be developed as original student research work under SLIIT's intellectual property policy. Upon commercialization, the team would seek to register the audit logging mechanism as a design innovation with the National Intellectual Property Office of Sri Lanka (NIPO). The codebase would be protected under a proprietary licence, with potential open-sourcing of non-core components to build community adoption.

7.BUDGET AND JUSTIFICATION

Item	Description	Estimated Cost
LLM API Usage (OpenAI - GPT-4o)	Required for medical intent parsing, RAG-based response generation, and testing clinical summaries.	LKR 6,000 - 8,000
Superbase (Search & Storage)	Hosting he Vector Database (Azure AI Search) and Blob Storage for patient report PDFs.	LKR 7000
Software Licences	Natural Language Processing and AI model processing	LKR 15000
Software Tools	Development tools, design software, testing tools	LKR 5000
Internet & Networking	Internet services used during development and testing	LKR 4000
Documentation & Printing	Report preparation and printing costs	LKR 3000
Miscellaneous	Contingency expenses	LKR 2000
TOTAL	LKR 45000-50000	

8. WORK BREAKDOWN STRUCTURE (WBS)

Main Module	Sub Task/ Feature
Project Initiation	• Literature review & feasibility assessment • Requirements specification & system design • Define module scope, roles & access levels
Database & Schema Design	• Design ERD: Patient, Doctor, Report, AuditLog, ClinicalNote tables • Define foreign keys, relationships & constraints • Set up PostgreSQL database & migration scripts
Patient Profile Module	• Build patient registration form with validation • Unique Patient ID generation (PAT-2025-001) • Edit, archive & search patient profiles • Link patients to assigned doctors & reports
Clinical Record Management	• File upload system (PDF, JPEG, DICOM) • Categorise by type: Lab, Radiology, Pharmacy, Clinical • Record listing with filters (date, type, doctor)

	• Link uploaded records to patient profiles
Audit & Change Tracking (Novelty)	• Implement middleware interceptor for all CRUD actions • Log: Doctor ID, action type, field, old→new value, UTC timestamp • Build immutable AuditLog table (append-only) • Visual audit trail timeline UI per patient • Admin dashboard: system-wide audit & compliance view
Clinical Notes & Versioning	• Add structured clinical notes editor per patient • Save each edit as a new version (append, not overwrite) • Diff-style version history viewer (what changed)
Access Control & Security	• Role-based record visibility (Junior / Consultant / Admin) • Flag sensitive records (mental health, HIV, confidential) • Patient consent tracking & permission guard
Testing & Documentation	• Unit & integration testing of audit log accuracy • Test role-based access restrictions (UAT) • PDF export: patient record summary & audit report • Final documentation & presentation preparation

9.GANTT CHART

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
Requirements & DB Design	■	■												
Database Setup			■											
REST API – CRUD				■	■									
Audit Log Engine					■	■								
RBAC Middleware						■	■							
Frontend – Dashboard								■	■					
Audit Trail UI									■					
Soft Delete & Recovery									■					
Completeness Scorer										■				
Integration (AI/Reports)											■			
Testing & Evaluation												■		

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
Documentation & Report													■■	■■

10. REFERENCES LIST

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4. [4] K. Cresswell, A. Callaghan, and A. Sheikh, "The NHS national programme for IT: A case study of programme management," International Journal of Medical Informatics, vol. 88, pp. 94–103, 2016.
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11.APPENDICES

Appendix A – Survey Instruments

Purpose: To collect user requirements from doctors and healthcare staff for the system.

Section 1 – Respondent Information

1. What is your current role? ○ Junior Doctor ○ Consultant ○ Administrator
2. Years of experience in healthcare: _____

Section 2 – Current Challenges

3. How much time do you spend interpreting a single medical report?
   ○ <5 min ○ 5–15 min ○ 15–30 min ○ >30 min
4. How often do you find medical terminology difficult to interpret?
   ○ Never ○ Rarely ○ Sometimes ○ Often ○ Always
5. Do you currently use a digital system for patient records? ○ Yes ○ No

Section 3 – Feature Preferences

6. Which features are most important to you? (Select all that apply)
   - Automated report summary
   - Abnormal value highlighting
   - AI chatbot for medical queries
   - Role-based access control
   - Immutable audit trail
   - Data completeness scoring

Appendix B – Detailed Diagrams

B.1 High-Level Architecture

React.js frontend → Node.js + Express.js REST API → PostgreSQL database, with external LLM/AI API integration. All communication is HTTPS/TLS encrypted; JWT authentication is enforced at the API gateway.

B.2 Data Flow Summary

(1) Doctor authenticates via JWT → (2) Access/create patient profile → (3) Upload medical report → (4) AI engine analyses report → (5) Structured summary stored → (6) All changes recorded in immutable AuditLog.

Appendix C – Preliminary UI Mockups

Mockups were designed in Figma during the requirements phase. Key screens:

C.1 Doctor Login Screen

Email/password fields, medical license verification badge, role-based redirect after login.

C.2 Patient Profile Dashboard

Demographic card, data completeness score (progress bar), tabs: Overview | Clinical Notes | Medical Reports | Audit Trail. Edit button visible to Consultants/Admins only.

C.3 Audit Trail / Clinical Timeline

Chronological log of all events showing event type, clinician, timestamp, and field-level changes (old value → new value). Filterable by date, event type, and user.

C.4 Medical Report Upload & AI Analysis

Drag-and-drop upload zone (PDF/image), AI result panel with report summary, abnormal value flags (red), key terms identified, and disclaimer: “This analysis does not substitute clinical judgment.”

C.5 AI Chatbot Interface

Chat window with medical-context prompting, doctor query input, AI response with report citations, and persistent disclaimer banner.

Appendix D – Risk Analysis

Risk	Likelihood	Impact	Mitigation
Real patient data exposure	Low	High	Use synthetic data only; no live clinical deployment
AI API downtime	Medium	High	Fallback static summary mode; cache recent responses
Scope creep	High	Medium	Strict WBS adherence; change request log maintained
Database failure / data loss	Low	High	Automated PostgreSQL backups; cloud-managed DB service
Unauthorized record access	Low	Critical	JWT + RBAC + HTTPS/TLS + AES-256 encryption at rest
Performance under load	Medium	Medium	Load test at 100 concurrent users; DB indexing applied.
Inaccurate AI output	Medium	High	Clear disclaimers; validate against curated test cases.
Schedule delays	Medium	Medium	Buffer weeks in Gantt chart; bi-weekly supervisor reviews.